

Zhi Ming Xu

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EDUCATION

Columbia University | New York, NY

Diploma Expected May 2028

- Bachelor of Science in Computer Engineering
- Cumulative GPA: 3.93/4.0; Dean's List
- Coursework: Signals and Systems, Data Structures & Algorithms, Fundamentals of Computer Systems, Electronic Circuits, Advanced Programming, Circuit Analysis, Digital VLSI Circuits, Computer Architecture, Advanced Logic Design

SKILLS

Languages: Python, C, C++, Java, TypeScript, SystemVerilog, VHDL, Verilog, Bash, MATLAB, HTML, CSS

Frameworks & Platforms: Node.js, Fastify, LangGraph, LangChain, Langfuse, Java Spring, Elasticsearch/OpenSearch, Instana, Claude Code

Tools & Cloud Infrastructure: Docker, Kubernetes, Git, GitHub Actions, Jenkins CI/CD, CMake, GNU Make, Xilinx Vivado, STM32CubeIDE

Hardware & Protocols: STM32 (ARM Cortex-M), FPGA, Model Context Protocol (MCP), UART, I2C, SPI, PWM, Single-Wire, GPIO

EXPERIENCE

IBM Corporation | Software Engineering Intern

May 2026 – Aug 2026

- Engineered a stateful multi-agent AIOps diagnostic platform in TypeScript, Node.js, and LangGraph; orchestrated 3 parallel worker nodes via Model Context Protocol (MCP) servers to automate Kubernetes, OpenSearch, and Instana log correlation.
- Automated root-cause analysis across distributed Kubernetes clusters and data centers, synthesizing up to 41K tokens of raw log dumps and system metrics to replace a 30-minute manual triage process with an automated 3-minute SLA.
- Constructed an automated test pipeline featuring offline mock data to validate graph routing and scheduled LLM integration tests with schema diffs for reasoning quality, containerizing the agent with Docker and Jenkins CI/CD to build images on code changes.

Columbia MobileX Lab | Undergraduate Research Assistant

Jan 2026 – May 2026

- Co-developed embedded C++ firmware on Arduino with PhD researchers for an AI voice anti-spoofing platform, implementing a deterministic state machine and 10s initialization sequence to eliminate transient noise and achieve 100% signal stability.
- Configured an embedded Linux audio node on Raspberry Pi, writing Python automation scripts to interface with ALSA drivers for automated 24-bit/48kHz playback testing across 100+ AI voice samples.

Columbia Billinge Group | Undergraduate Research Assistant

May 2025 – Dec 2025

- Optimized algorithmic compute bottlenecks within open-source scientific computing packages (diffpy.pdfgetx, diffpy.xpdfsuite), achieving a 40% reduction in data processing latency via vectorization and improved memory management.
- Designed and maintained an automated DevOps CI/CD pipeline using GitHub Actions and pre-commit automation across 5+ distributed packages; integrated pytest-based functional testing to sustain 90%+ test coverage and guarantee code stability.

Save Aways | Software Engineering Intern

July 2025 – Aug 2025

- Deployed backend recommendation services in Java Spring and Elasticsearch, reducing search query latency by 25% through indexing optimization; integrated Stripe and PayPal REST APIs, reducing transaction errors by 15%.

PROJECTS

FPGA Digital Audio Synthesizer | Xilinx Artix-7, VHDL, Vivado

github.com/zmx27/FPGA-Audio-Synthesizer

- Synthesized a digital audio core deployed on a Xilinx Artix-7 FPGA using VHDL and Xilinx Vivado; implemented cascaded clock dividers to derive musical frequencies from a 100 MHz system clock with sub-1% frequency error.
- Built a custom Finite State Machine (FSM) and ROM-based lookup tables (LUTs) to drive an automated hardware playback engine, multiplexing 32-beat sequential digital instructions in real-time.
- Conducted functional verification via VHDL testbenches to perform waveform-level timing analysis and simulation; built a time-multiplexed 4-digit 7-segment display controller module scanning at 10 kHz to eliminate UI rendering overhead.

Embedded Environmental Monitor | STM32, C, I2C

github.com/zmx27/STM32-as-Sensor

- Developed an embedded environmental monitor built around an ARM Cortex-M4 (STM32F401RE) MCU integrated with a DHT11 temperature sensor to process localized telemetry using optimized bare-metal firmware.
- Authored a custom bare-metal single-wire communication driver in C, implementing microsecond-precision hardware timer delays to handle bit-banging and accurately parse a 40-bit sensor data stream.
- Implemented an I2C peripheral driver operating at a 100 kHz clock frequency to interface a 16x2 character LCD for data rendering.